

SHARAN GANESHBABU

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SUMMARY

Innovative and results-driven Software Engineer with extensive experience designing and implementing enterprise-scale solutions across diverse industries. Skilled at developing distributed applications, optimizing data pipelines, and integrating modern cloud-native platforms to support large-scale business operations. Proven ability to deliver robust, high-performance systems through strong coding practices, automation, and cross-functional collaboration. Adept at bridging technical and business needs to deliver scalable, reliable, and efficient solutions.

TECHNICAL SKILLS

Programming & Frameworks: Java 11, Scala, Python, Spring Boot, REST APIs, Microservices, Looper

Big Data & Streaming: Apache Spark (Java/Scala), Apache Kafka (streaming & processing), Hive, Cassandra

Cloud Platforms & Databases: Google Cloud Platform (GCP – BigQuery, Pub/Sub, Cloud Storage), Oracle, SQL

CI/CD & DevOps: Concord CI/CD, Kubernetes, Docker, Git (branching, code reviews, pipelines)

Workflow & Automation: Apache Airflow, ETL Pipelines, Workflow Orchestration, Automation Scripts

Methodologies & Practices: Agile/Scrum, Test-Driven Development (TDD), Scalable Systems Design, Performance Tuning

Other Tools: Tableau, Power BI (for visualization), ServiceNow, Salesforce

PROFESSIONAL EXPERIENCES

Software Engineer – ReliantHeart Inc., Houston, TX

January 2025 – Present

- Designed and implemented **Spark with Java** pipelines for large healthcare datasets, improving ETL throughput by 60% while maintaining strict HIPAA compliance.
- Built **Spring Boot microservices** to support modular medical applications, reducing system coupling and accelerating release cycles by 30%.
- Deployed **Kafka clusters** for streaming IoT telemetry data from patient devices, ensuring real-time monitoring and reducing latency in alert generation.
- Leveraged **GCP services** (BigQuery, Pub/Sub, Cloud Storage) to build cloud-native analytics platforms that improved scalability and reduced infrastructure costs.
- Integrated **BigQuery** for high-speed querying of medical records, reducing reporting turnaround by 40%.
- Developed **Hive-based analytical tables** to unify structured and semi-structured patient data for predictive modeling.
- Automated deployments using **Concord CI/CD pipelines**, reducing manual errors and enabling continuous delivery.
- Managed distributed systems with **Kubernetes**, scaling workloads dynamically for fluctuating patient monitoring demand.
- Standardized branching strategies in **Git**, improving traceability and developer collaboration across global teams.
- Built orchestration pipelines with **Airflow** to automate reporting, compliance checks, and data validation workflows.
- Extended pipelines with **Scala** and **Python**, enhancing Spark performance tuning and enabling custom ML integrations.
- Implemented **Cassandra clusters** to support high-velocity patient monitoring data, ensuring low-latency access.
- Migrated clinical data to **Oracle**-based systems for compliance reporting and auditing.
- Used **Looper** for resilient background job processing tied to Kafka event streams, ensuring reliable task execution.
- Optimized **system observability and monitoring** using logs and metrics, reducing incident resolution time by 25%.
- Partnered with **cross-functional clinical teams** to translate patient requirements into technical workflows with measurable outcomes.

Environment: Spark (Java/Scala), Spring Boot, Kafka, GCP (BigQuery, Pub/Sub, Cloud Storage), Hive, Concord CI/CD, Kubernetes, Git, Airflow, Python, Cassandra, Oracle, Looper, Microservices.

Software Engineer – University of Houston, Houston, TX

January 2023 – December 2024

- Engineered **Spark with Java** ETL jobs for exam scheduling data, reducing processing times for millions of records.
- Deployed **Spring Boot APIs** to support exam scheduling, student records, and automated reporting systems.
- Implemented **Kafka streaming systems** for real-time exam check-in and attendance tracking, reducing no-show rates.
- Migrated academic data pipelines to **GCP**, leveraging BigQuery and Pub/Sub for scalable analytics.
- Designed complex **BigQuery queries** to provide leadership insights into exam center utilization and trends.
- Integrated **Hive warehouses** for structured storage of historical exam data, improving auditing and compliance.
- Automated application delivery through **Concord CI/CD**, reducing lead time for deployments.
- Scaled exam-processing services with **Kubernetes**, ensuring zero downtime during peak exam periods.
- Managed **Git repositories** with code reviews, automated testing, and branch protections to enforce quality standards.
- Developed **Airflow DAGs** to orchestrate ETL, scheduling, and audit-reporting workflows.
- Used **Scala** for Spark UDFs that supported advanced exam analytics and scheduling algorithms.
- Automated data validation and reporting in **Python**, reducing manual reconciliation by 70%.
- Leveraged **Cassandra** for storing student interactions and logs with millisecond-level query speeds.
- Integrated data with **Oracle systems** for compliance with accreditation and regulatory reporting.
- Adopted **Looper** for background exam-notification tasks tied to Kafka events, ensuring timely alerts.
- Enhanced **security and access controls** for sensitive exam data, ensuring compliance with academic standards.
- Partnered with **academic leadership** to deliver automated dashboards that drove 40% improvements in resource allocation.

Environment: Spark (Java/Scala), Spring Boot, Kafka, GCP (BigQuery, Pub/Sub, Cloud Storage), Hive, Concord CI/CD, Kubernetes, Git, Airflow, Python, Cassandra, Oracle, Looper.

- Built **Spark with Java** frameworks to process 5M+ daily transactions, improving fraud detection by 25%.
- Developed **Spring Boot APIs** for customer transaction services, improving modularity and performance.
- Integrated **Kafka** for high-volume event streaming across payment, risk, and fraud systems.
- Leveraged **GCP** for cloud migration of financial reporting pipelines, reducing infrastructure costs by 35%.
- Designed **BigQuery data marts** for executive dashboards, enabling real-time decision support.
- Used **Hive** for financial transaction aggregation and compliance reporting, reducing manual reconciliation efforts.
- Implemented **Concord CI/CD pipelines** to automate build, test, and release processes.
- Managed **Kubernetes clusters** hosting microservices across production and DR environments.
- Standardized codebase collaboration with **Git**, reducing integration issues by 40%.
- Automated transaction monitoring workflows using **Airflow DAGs**, ensuring real-time SLA compliance.
- Extended Spark jobs with **Scala**, enhancing streaming and batch processing performance.
- Applied **Python** for data cleansing, risk modeling, and ETL automation.
- Configured **Cassandra clusters** for scalable, fault-tolerant financial data storage.
- Migrated core reporting and audit systems to **Oracle**, ensuring regulatory compliance.
- Used **Looper** to manage asynchronous transaction reconciliation tasks, ensuring reliable execution.
- Conducted **capacity planning and performance tuning** of Kafka clusters to sustain peak trading loads.
- Collaborated with **regulatory auditors** to validate compliance frameworks built into transaction pipelines.

Environment: Spark (Java/Scala), Spring Boot, Kafka, GCP (BigQuery, Pub/Sub, Cloud Storage), Hive, Concord CI/CD, Kubernetes, Git, Airflow, Python, Cassandra, Oracle, Looper, Microservices.

EDUCATION

University of Houston, Houston, TX – *Master of Science in Computer and Systems Engineering*

Lovely Professional University, Punjab, India – *Bachelor of Technology in Mechatronics Engineering*

CERTIFICATIONS

- Microsoft Azure Data Fundamentals (DP-900)
- Palo Alto Networks Certified Network Security Engineer (PCNSE)
- AWS Certified Cloud Practitioner – *In Progress*